Network Based Intrusion Detection Using Dense Layers, Bi-LSTM with Multi-Head Attention and XGBoost Classification

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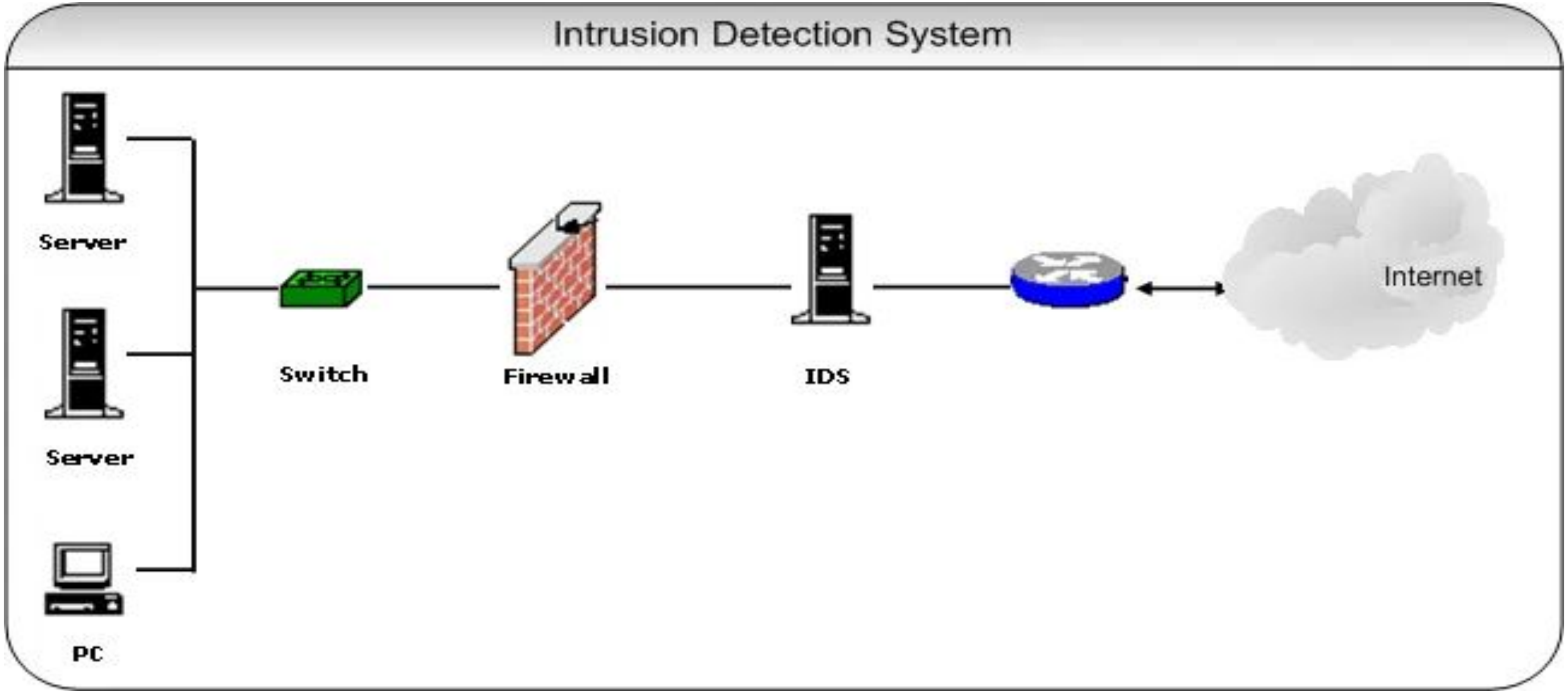
Introduction to Intrusion Detection System



Intrusion Detection System (IDS)

- An Intrusion Detection System (IDS) is a crucial component of network security, designed to monitor and analyze the traffic within a network in **real-time**[13].
- It identifies and alerts on any suspicious activities that could indicate a potential attack, breach, or unauthorized access attempt[8].

Importance of IDS

- **1. Increasing Network Complexity**: As networks become more interconnected with diverse devices (IoT, cloud infrastructure, etc.), the attack surface grows, providing more opportunities for cybercriminals to exploit vulnerabilities[2].
- **2. Rise in Cyberattacks**: Cyberattacks such as Distributed Denial of Service (DDoS), malware, phishing, and ransomware have become more frequent and sophisticated, targeting organizations of all sizes and industries[7].
- **4. Zero-Day Vulnerabilities**: One of the most critical threats today, zero-day attacks exploit unknown vulnerabilities for which no patch or fix is available. IDS, especially when integrated with AI, can detect abnormal behavior that signals such exploits[9].
- **5. Real-time Detection and Response**: Modern IDS solutions provide real-time monitoring and alerting, allowing organizations to respond swiftly to threats, minimizing potential damage[3].
- **6. Multi-layered Security**: IDS acts as a critical layer in an organization’s security infrastructure, complementing firewalls, antivirus software, and other security mechanisms to provide comprehensive protection[12].

Functionality of Intrusion Detection System (IDS)

- **Monitoring Network Traffic**: IDS continuously scans the flow of data between users and devices, evaluating network packets, identifying anomalies, and comparing behavior to established patterns of normal and malicious activity[11].
- **Real-time Analysis**: By instantly processing traffic data, IDS can detect potential intrusions as they occur, providing immediate alerts for further action[10].

Integration of AI in IDS

- **AI-Enhanced Detection**: Intrusion Detection Systems (IDS) leverage **Artificial Intelligence (AI)**[6].
- particularly **Machine Learning (ML)** and **Deep Learning (DL)**
- to improve the precision and efficiency of detecting both known and unknown security threats[9].

Role of Deep Learning (DL)

- **Neural Networks for Complex Patterns**: Deep learning, especially through techniques like **Convolutional Neural Networks (CNN)** and **Recurrent Neural Networks (RNN)** (including **BiLSTM**), excels at extracting complex, hierarchical features from vast amounts of network traffic data. This allows the system to recognize sophisticated attack patterns that may be missed by traditional methods.[6]
- **High Detection Accuracy**: DL models can process large-scale datasets, improving detection accuracy by identifying minute, previously undetectable correlations between network activities.[6]

Objective

- To introduce a **hybrid model** that integrates **Dense Layer** with **Bi-LSTM** for enhanced intrusion detection.
- To demonstrate how this model **learns complex patterns** in network traffic.
- To showcase the model's ability to **accurately detect intrusions** by analyzing both spatial and temporal data.
- To highlight the potential of **attention mechanisms** in improving focus on critical data segments.

Literature Survey

- Vanlalruata Hnamte et al, Jamal Hussain et al. (March 2023) DCNNBiLSTM: An Efficient Hybrid Deep Learning-Based Intrusion Detection System
  - The model was trained on **CICIDS2018** and **Edge_IIoT** datasets, achieving **100%** and **99.64%** accuracy, respectively.
  - It combines the CNN's ability to extract local features with BiLSTM's strength in capturing long-term dependencies in network traffic.
  - The hybrid model outperforms traditional models like **DNN, LSTM,** and **CNN** alone in terms of accuracy and loss rate.
  - The architecture includes CNN layers for feature extraction, BiLSTM for sequence prediction, and a fully connected layer for classification, with significant improvements in detecting **DDoS attacks** and other intrusions.

Literature Survey

- **Rachid Ben Said et al. (December 2023)**: *"CNN-BiLSTM: A Hybrid Deep Learning Approach for Network Intrusion Detection System in Software-Defined Networking with Hybrid Feature Selection"*
  - The model was trained on **NSL-KDD**, **UNSW-NB15**, and **InSDN** datasets, achieving the highest accuracy of **98.42%** on NSL-KDD.
  - Outperforms traditional models like **AlexNet, LeNet5, CNN-LSTM** in accuracy and training time.
  - The model effectively detects complex attacks, including **DDoS, U2R, R2L**, and **Web Attacks**, with high precision and recall.
  - The study also emphasizes using **feature selection** techniques like **Random Forest** and **Recursive Feature Elimination (RFE)** for enhanced performance.

Literature Survey

- **Sapna Sadhwani et al. (January 2024)**: *"BiLSTM-CNN Hybrid Intrusion Detection System for IoT Application"*
  - The model was trained on the **UNSW-NB15** dataset, achieving a precision of **99.26%**, accuracy of **97.51%**, and F1 score of **93.09%**.
  - It combines **BiLSTM** for temporal feature learning and **CNN** for spatial feature extraction to enhance the detection of IoT network intrusions.
  - The hybrid BiLSTM-CNN model outperformed individual BiLSTM and CNN models, demonstrating superior accuracy, false positive rate, and false negative rate.
  - The system was designed for detecting various attacks in IoT networks, including **DDoS**, **exploits**, and **reconnaissance** attacks

Literature Survey

- **Muhammad Ashfaq Khan et al. (May 2021)** HCRNNIDS: Hybrid Convolutional Recurrent Neural Network-Based Network Intrusion Detection System.
  - The proposed model was trained on the **CSE-CIC-IDS2018** dataset and achieved an accuracy of **97.75%.**
  - HCRNNIDS combines **CNN for spatial feature extraction and RNN** for **capturing temporal dependencies**, enhancing the model's performance in detecting network intrusions.
  - The hybrid approach outperforms traditional machine learning models by addressing class imbalance issues and improving detection rates for attacks like **DDoS, brute-force,** and **infiltration**.

Literature Survey

- Yakubu Imrana et al, Yanping Xiang et al, Liaqat Ali et al, Zaharawu Abdul-Rauf et al. (June 2021) Bidirectional LSTM deep learning approach for intrusion detection
    - The **NSL-KDD dataset** is used to train and evaluate the model.
    - **BiLSTM outperforms conventional LSTM** models in detecting intrusions, especially for hard-to-detect attacks like **User-to-Root (U2R)** and **Remote-to-Local (R2L)**.
    - The proposed BiLSTM model achieves a **higher accuracy**, **precision**, **recall**, and **F-score**, with significantly **lower false alarm rates** than other state-of-the-art techniques.
    - It shows an improvement of **7.59% and 4.45%** in detection accuracy over conventional LSTM for KDDTest−21 and KDDTest+ datasets, respectively.

Literature Survey

- Azriel Henry et al, Sunil Gautam et al, Samrat Khanna et al, Khaled Rabie et al, Thokozani Shongwe et al,Pronaya Bhattacharya et al, Bhisham Sharma et al and Subrata Chowdhury et al. (January 2023) Composition of Hybrid Deep Learning Model and Feature Optimization for Intrusion Detection System
  - The model achieved an accuracy of **98.73%** and a **False Positive Rate (FPR)** of **0.075**.
  - They compared CNN-GRU to other methods and found it more effective for detecting attacks like SQL injection and SSH Patator but **failed to detect Heartbleed** and XSS attacks.
  - Their approach optimized the dataset by reducing redundant features and utilizing **Pearson's Correlation Coefficient** for feature selection.

	Reference	Existing Technique	Summary	Pros	Cons
1	Vanlalruata Hnamte et al., (2023)	DCNN-BiLSTM Hybrid IDS	Classifier: CNN, BiLSTM; Dataset: CICIDS2018, Edge_IIoT; Improved DDoS detection, hybrid model outperforms traditional models.	High accuracy, handles complex intrusions, effective DDoS detection	May require high computational resources, potential overfitting on small datasets
2	Rachid Ben Said et al., (2023)	CNN-BiLSTM with Hybrid Feature Selection	Classifier: CNN, BiLSTM; Dataset: NSL-KDD, UNSW-NB15, InSDN; Effective for DDoS, U2R, R2L attacks with Random Forest and RFE feature selection.	Hybrid approach enhances performance, suitable for a variety of attack types	Relatively lower accuracy on more recent datasets, longer training time due to hybrid model
3	Sapna Sadhwani et al., (2024)	BiLSTM-CNN Hybrid IDS for IoT	Classifier: BiLSTM, CNN; Dataset: UNSW-NB15; Outperformed individual models for IoT intrusion detection.	Excellent precision and accuracy for IoT networks	May struggle with real-time detection on larger datasets, prone to high memory usage

4	Muhammad Ashfaq Khan et al., (2021)	HCRNNIDS: Hybrid Convolutional Recurrent NIDS	Classifier: CNN, RNN; Dataset: CSE-CIC-IDS2018; Addresses class imbalance, effective for DDoS, brute-force, infiltration attacks.	Resolves class imbalance, strong against brute-force attacks	Limited scalability, longer training time due to complex model structure
5	Azriel Henry et al., Sunil Gautam, (2023)	CNN-GRU Hybrid IDS with Feature Optimization	Classifier: CNN, GRU; Dataset: CICIDS-2017; Combines CNN and GRU, optimizing dataset using Pearson's Correlation to reduce redundant features	High accuracy, effective feature selection, detects SQL Injection and SSH Patator attacks	Fails to detect Heartbleed and XSS attacks, requires large computational resources
6	Yakubu Imrana et al., (2021)	CNN-GRU with Feature Optimization	Classifier: CNN, GRU; Dataset: NSL-KDD and UNSW-NB15; Effective for SQL injection, SSH attacks; Pearson’s Correlation for feature selection.	Effective for feature selection, strong performance on common attacks	Struggles with advanced attacks like XSS, Heartbleed; may require more dataset diversity

7	Mohammed Jouhari et al. (2024)	Lightweight CNN-BiLSTM	Classifier: Lightweight CNN combined with Bidirectional LSTM; Dataset: UNSW-NB15; Achieved 97.28% accuracy for binary classification and 96.91% for multi-classification	Resolves class imbalance, strong against brute-force attacks	Limited scalability, longer training time due to complex model structure
8	FatimaEzzahra Laghrissi et al., (2021)	LSTM with PCA & Mutual Information	Classifier: LSTM; Dataset: KDD99; Used PCA and Mutual Information for dimensionality reduction;	PCA reduces computational cost; improves accuracy by removing irrelevant features.	Performance drops in multiclass classification due to noisy data; time-intensive without reduction.
9	D. V. Jeyanthi et al., (2023)	RNN-BiLSTM for IoT-based Healthcare IDS	Classifier: RNN-BiLSTM; Dataset: IoTID20; Custom features specific to healthcare, such as vital signs monitoring via IoT devices.	Highly effective for IoT-based healthcare environments, real-time detection.	Custom feature extraction specific to healthcare; not easily generalizable to other domains.

10	T. Thilagam and R. Aruna (2021)	Custom RC-NN with ALO Optimization	Classifier: RC-NN hybrid with CNN and LSTM; Used Ant Lion Optimization (ALO) to reduce error rate; Dataset: DARPA and CSE-CIC-IDS2018 datasets;	High classification accuracy and reduced error rate (0.0012); Efficient in detecting network-based cloud intrusions	High computational cost; Requires extensive training time
11	FatimaEzzahra Laghrissi et al. (2021)	LSTM with PCA & Mutual Information	Classifier: LSTM; Dataset: KDD99; Applied PCA for dimensionality reduction and Mutual Information; Best accuracy with PCA (99.49%) for binary classification	PCA reduces computational cost and improves accuracy by removing irrelevant features	Performance drops in multiclass classification due to noisy data; Time-intensive without reduction
12	Zhenxiang He et al. (2024)	SSAE-TCN-BiLSTM	Classifier: Stacked Sparse Autoencoder with Temporal Convolutional Network and Bidirectional LSTM; Dataset: UNSW-NB15, CICDS2017;	Excellent temporal data handling; High accuracy and F1 scores	Requires significant computational resources for training

13	Abdoul-Aziz Maiga et al. (2023)	LSTM+BiGRU+Bi LSTM	Classifier: Hybrid RNN (LSTM, Bidirectional GRU, Bidirectional LSTM) using Federated Learning for DDoS detection; Dataset: CICDDoS2019;	High detection accuracy with efficient federated learning; Ensures privacy	Complexity in model training due to multiple RNN layers
14	Syed Muhammad Salman Bukhari et al. (2024)	SCNN-BiLSTM	Classifier: Stacked CNN and Bidirectional LSTM using Federated Learning; Dataset: CIC-IDS2017, WSN-DS;	High accuracy in detecting unknown threats; Federated learning ensures data privac	Higher computational cost and complexity due to model architecture
15	Nuruzzaman Faruqui et al. (2023)	CNN-LSTM Hybridization	Classifier: CNN-LSTM hybrid for IoMT intrusion detection; Combines image-based and sequential data analysis; Dataset:Custom IoMT Dataset Designed to detect 12 different types of intrusions	Effectively handles both image and non-image data; Optimized for Detection Rate and False Positive Rate	Increased complexity due to multi-data handling; Potential processing delays for real-time IoMT devices

Research Gap

1. **Limited Feature Extraction**: Previous models using Bi-LSTM lacked spatial feature extraction, which is enhanced by integrating Dense Layer in the current hybrid model.
2. **Ineffective for Complex Attacks**: Older models struggled with attacks like SQL injection and Heartbleed, while the Dense Layer-BiLSTM hybrid improves detection of such intricate patterns.
3. **No Attention Mechanism**: Earlier approaches lacked attention layers, which are now used to focus on key data segments, improving detection accuracy and reducing false positives.
4. **High False Positives**: Previous models, like GRNN, had higher false positives; the hybrid model's Dense Layer and attention layer mitigate this issue.
5. **Poor Adaptability to New Attacks**: Traditional models were less effective against evolving threats, whereas the hybrid approach is more adaptable to zero-day attacks due to its deeper learning architecture.

Proposed Model

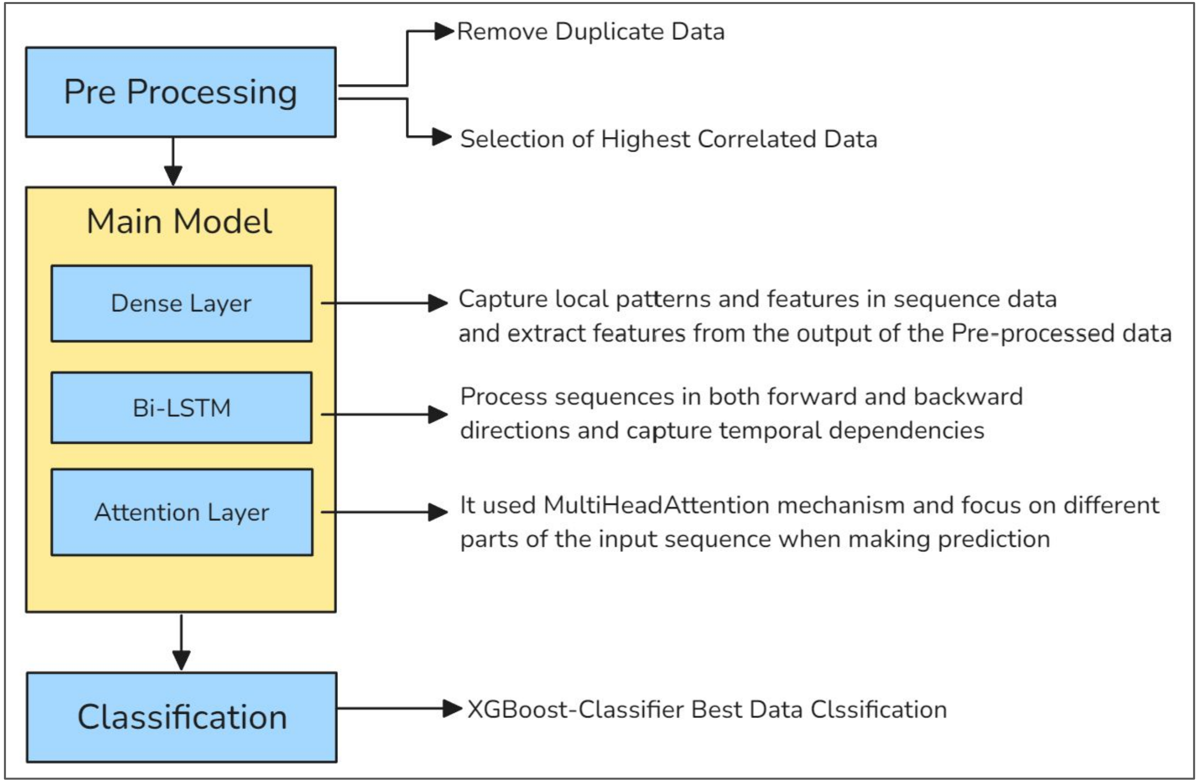


Figure 1: Proposed Model with 3 Stages 1. Data Preprocessing 2.Classification 3.Main Model

Proposed Model

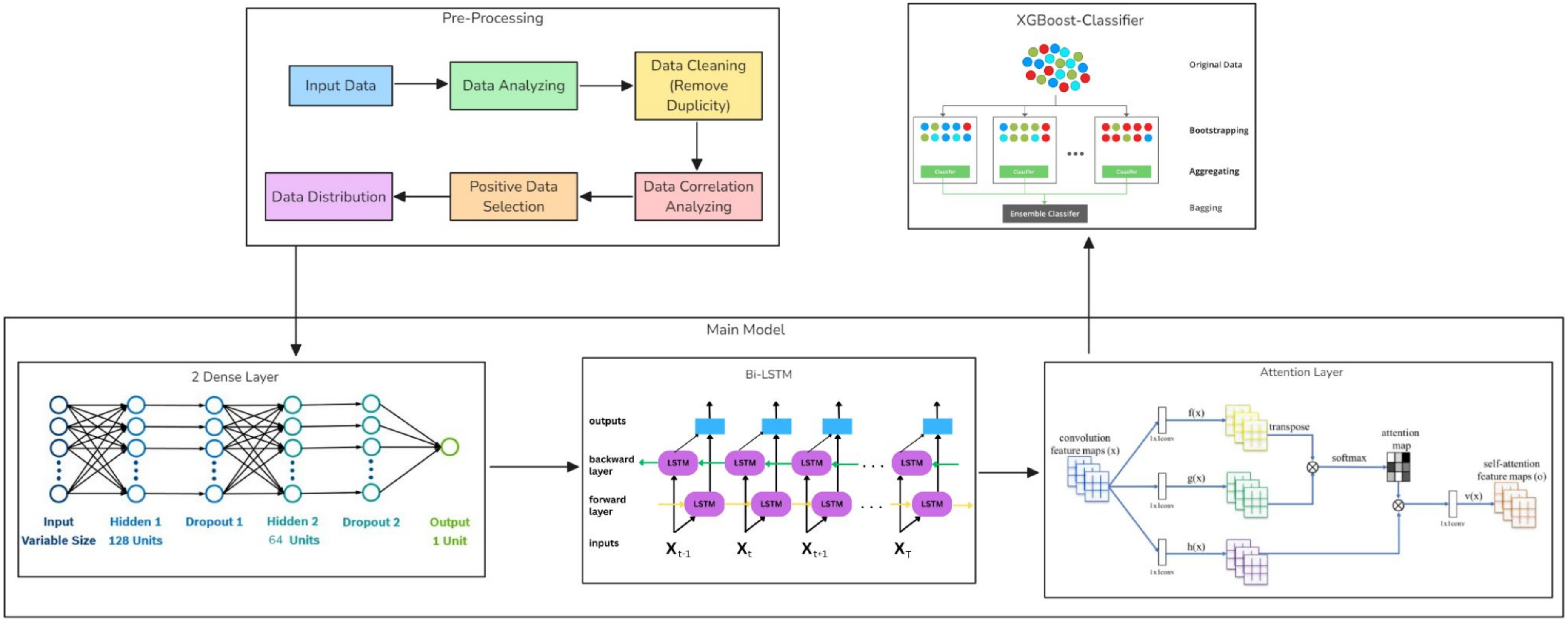


Figure 2: Proposed Model with 3 Stages 1. Data Preprocessing 2.Main Model 3.Classification

Algorithm : Proposed Model

Algorithm 1 Feature Optimization and Machine Learning Classification

Require: Dataset D with features X and target labels y

Ensure: Classification report and Confusion matrix

Data Loading and Preprocessing:
Load dataset D into pandas DataFrame
Rename target variable to ClassLabel
Split dataset: $X \leftarrow D \setminus \{\text{ClassLabel}\}$, $y \leftarrow D[\text{ClassLabel}]$
Standardize features: $X \leftarrow \text{StandardScaler}(X)$

Dataset Split:
$(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}}) \leftarrow \text{train\_test\_split}(X, y, 0.2)$

Model Architecture:
Create custom Attention Layer using Multi-Head Attention
Define feature extraction model:
if Branch 1 **then**
    Dense layers for feature learning
else if Branch 2 **then**
    Bidirectional LSTM with Attention mechanism
end if

Model Compilation:
optimizer $\leftarrow$ Adam(learning_rate $= 10^{-3}$)
loss $\leftarrow$ sparse_categorical_crossentropy
metrics $\leftarrow$ [accuracy]

Model Training:
Train model on X_{train} and y_{train}

Feature Extraction:
$X_{\text{train\_features}} \leftarrow \text{feature\_extractor}(X_{\text{train}})$
$X_{\text{test\_features}} \leftarrow \text{feature\_extractor}(X_{\text{test}})$

Classifier Training:
Train XGBoost classifier on $X_{\text{train\_features}}$ and y_{train}

Prediction:
$y_{\text{pred}} \leftarrow \text{xgb\_model}(X_{\text{test\_features}})$

Evaluation:
Generate classification report
Compute and display confusion matrix
    return Classification report, Confusion matrix

Dataset Information

- Four key datasets: **CIC-IDS2017**, **CIC-DoS2017**, **CSE-CIC-IDS2018**, and **CIC-DDoS2019** are consolidated into one collection.
- **Purpose**:
- Designed to provide a **larger, more diverse dataset** for Network Intrusion Detection Systems (NIDS) research and analysis.
- **Dataset Features**:
- **Cleaned and Harmonized**: Labels have been standardized, and flawed features identified in two studies have been removed.
- **Unified for Ease**:
- Though researchers often use individual datasets, this combined set is **compatible out-of-the-box** and processed with the same feature extraction tools.
- **Credits**:
- If using this combined dataset, **credit the original authors**. Citations are provided both on Kaggle and the CIC's official dataset page.

Dataset Features

1. Flow Duration
2. Flow Bytes/s & Packets/s
3. Total Fwd & Bwd Packets
4. Packet Length Mean/Max/Min
5. PSH & ACK Flag Count
6. Protocol Type
7. Bwd & Fwd IAT Mean/Std
8. Active & Idle Mean
9. Label
10. SYN Flag Count

Results and Discussion

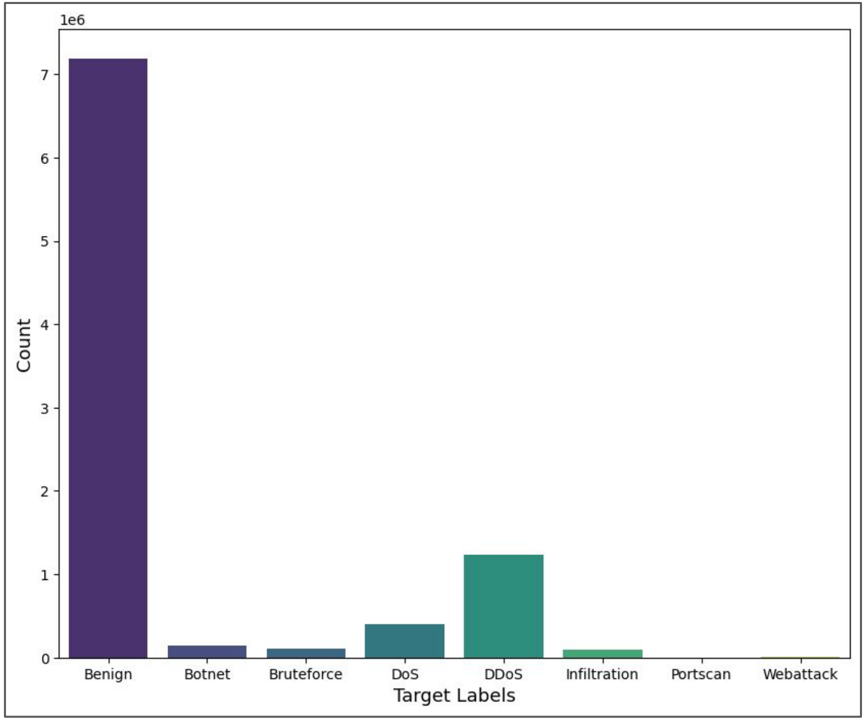


Figure 3: Target Distribution After Drop

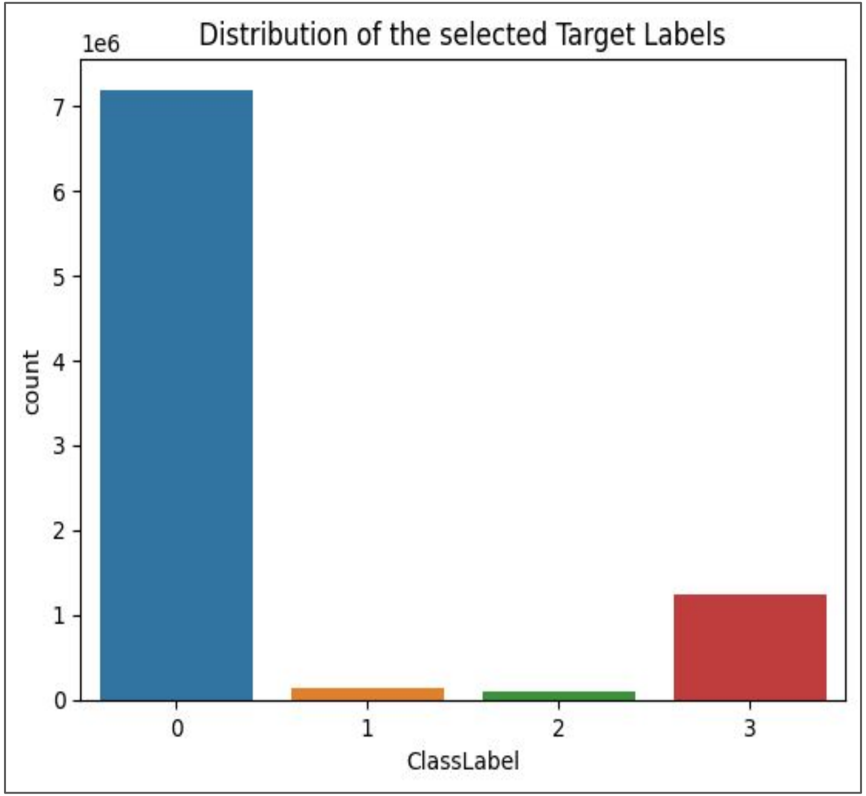


Figure 4: Distribution Chart of Selected Target Labels

Results and Discussion

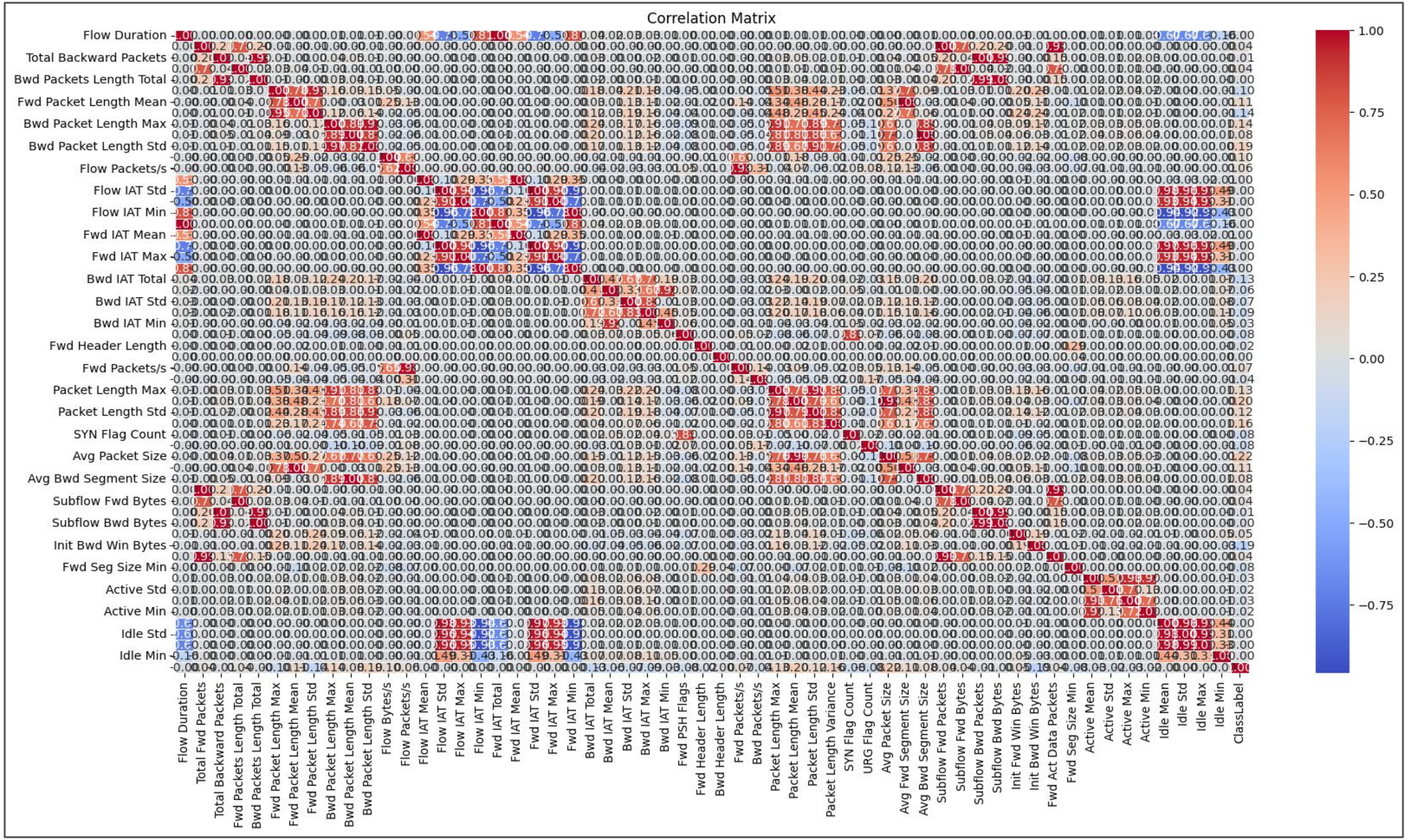
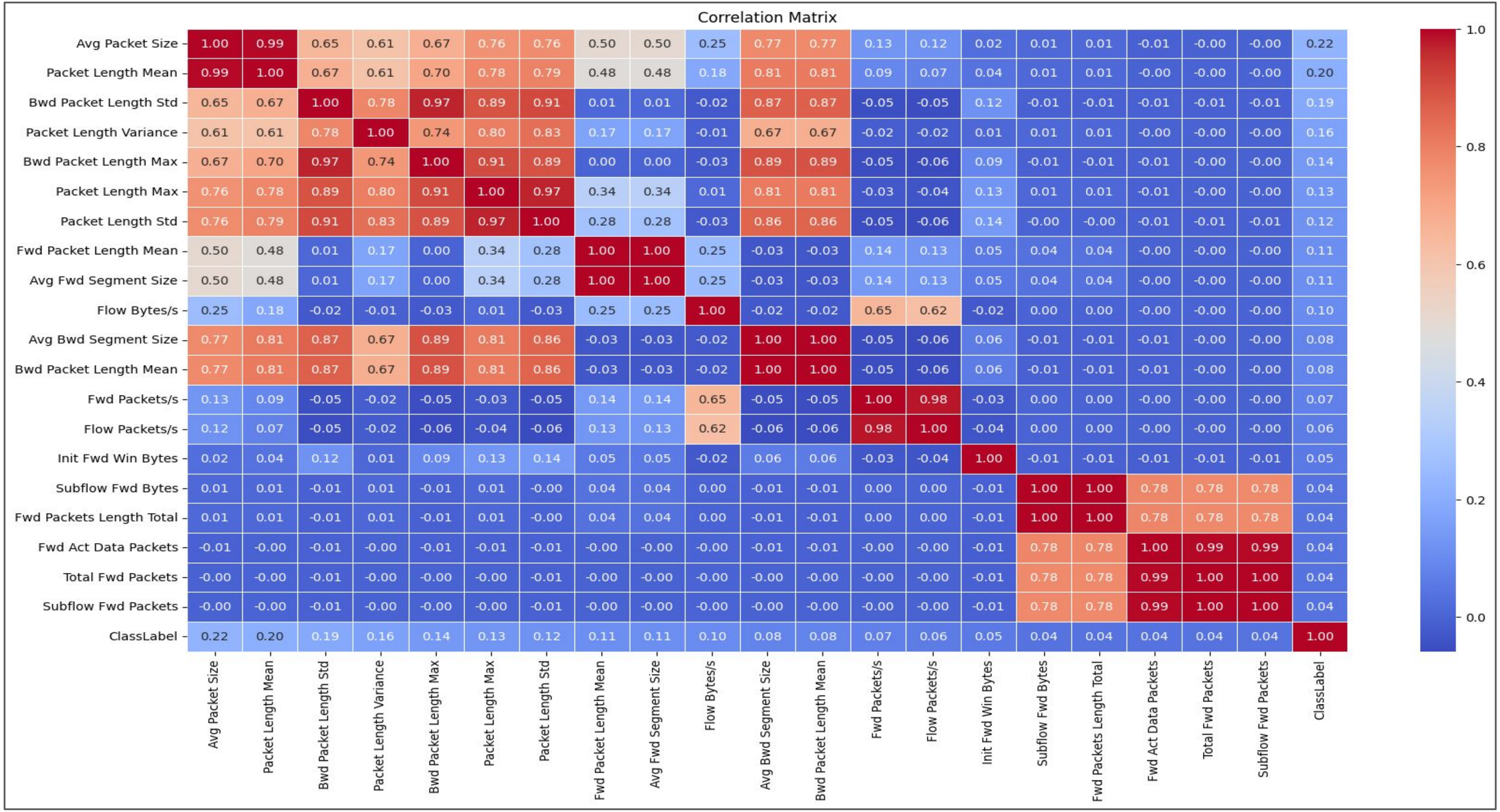


Figure 5: Correlation Matrix before preprocessing

Results and Discussion



Results and Discussion

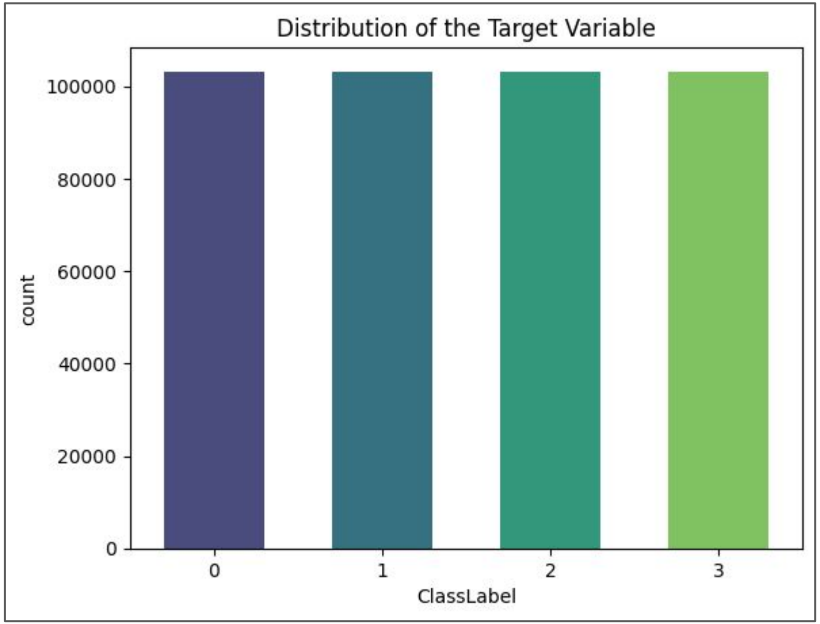


Figure 7: Distribution Chart of Target Variable

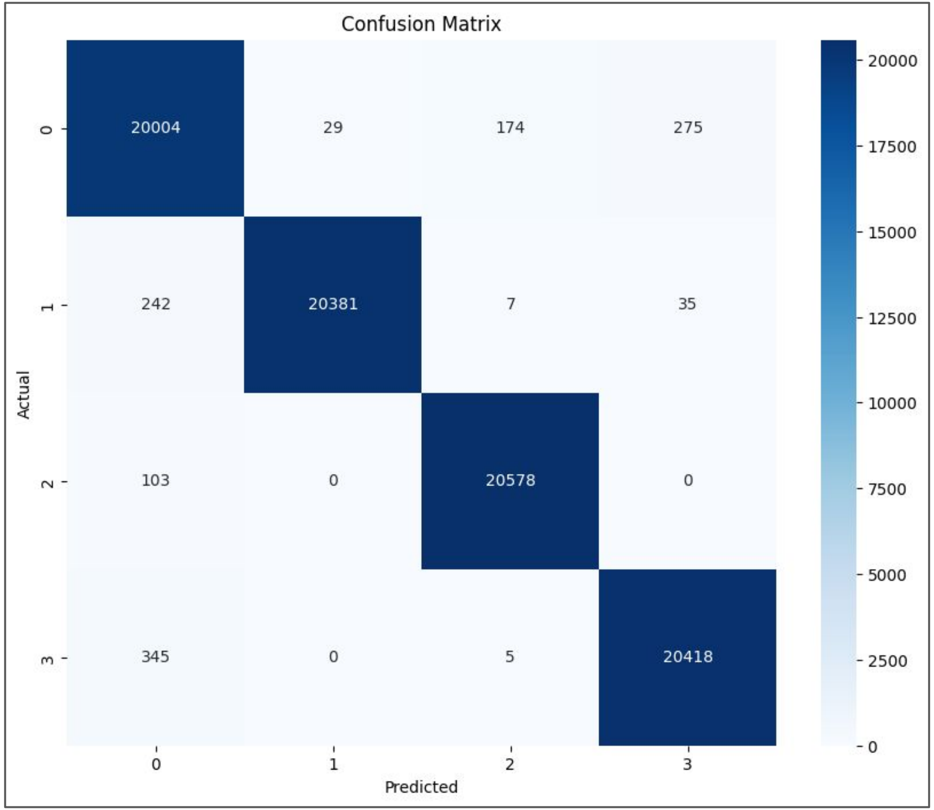


Figure 8: Confusion Matrix

Results and Discussion

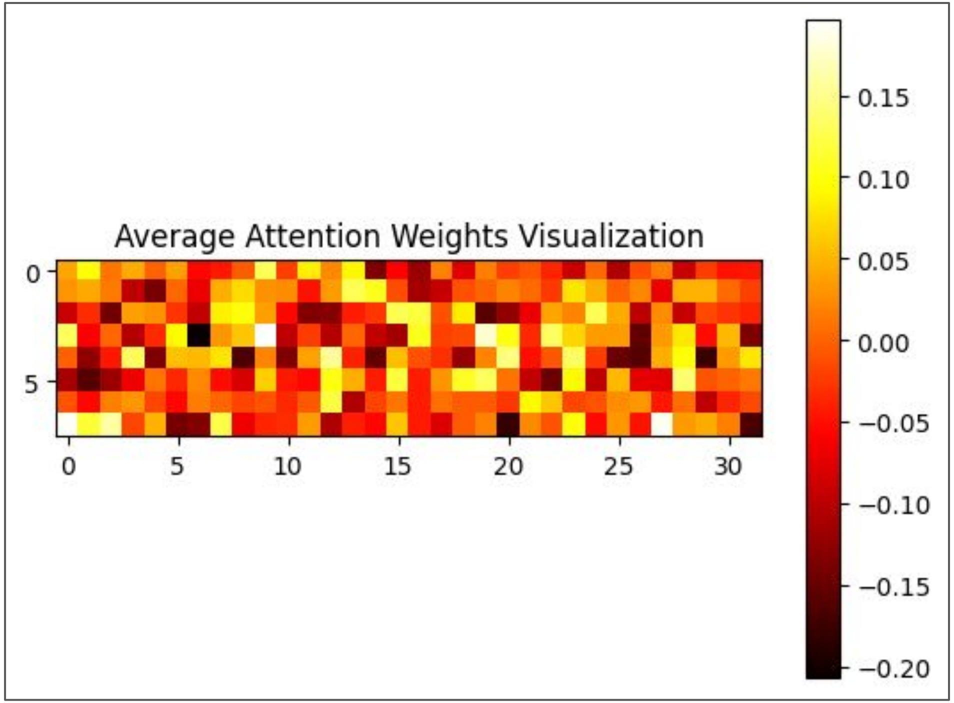


Figure 9: Average Attention Weights Visualization

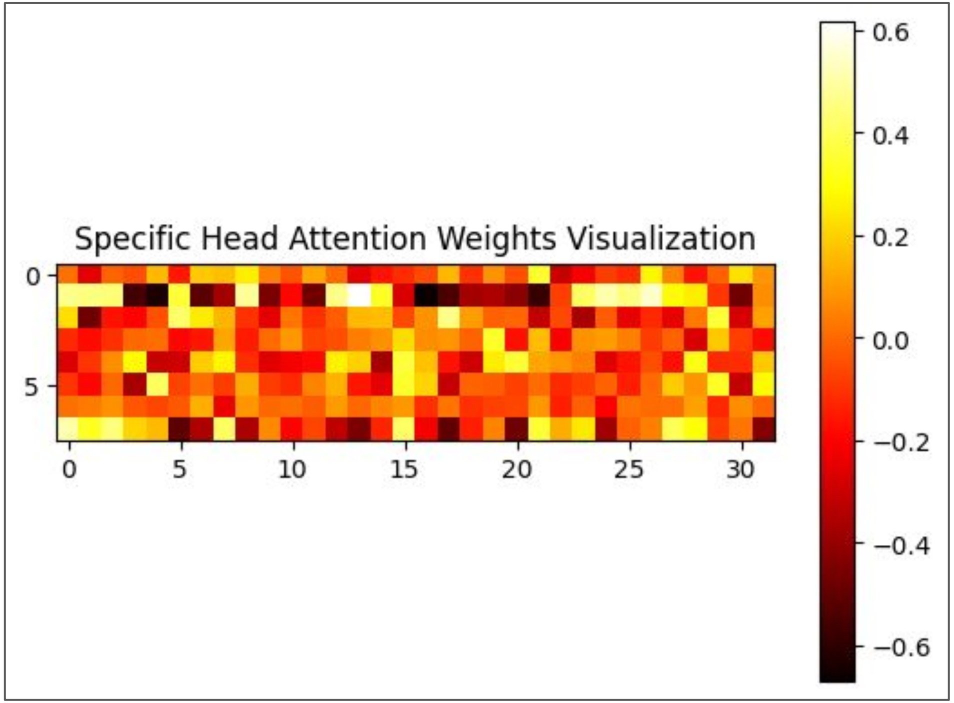


Figure 10: Specific Head Attention Weights Visualization

Results and Discussion

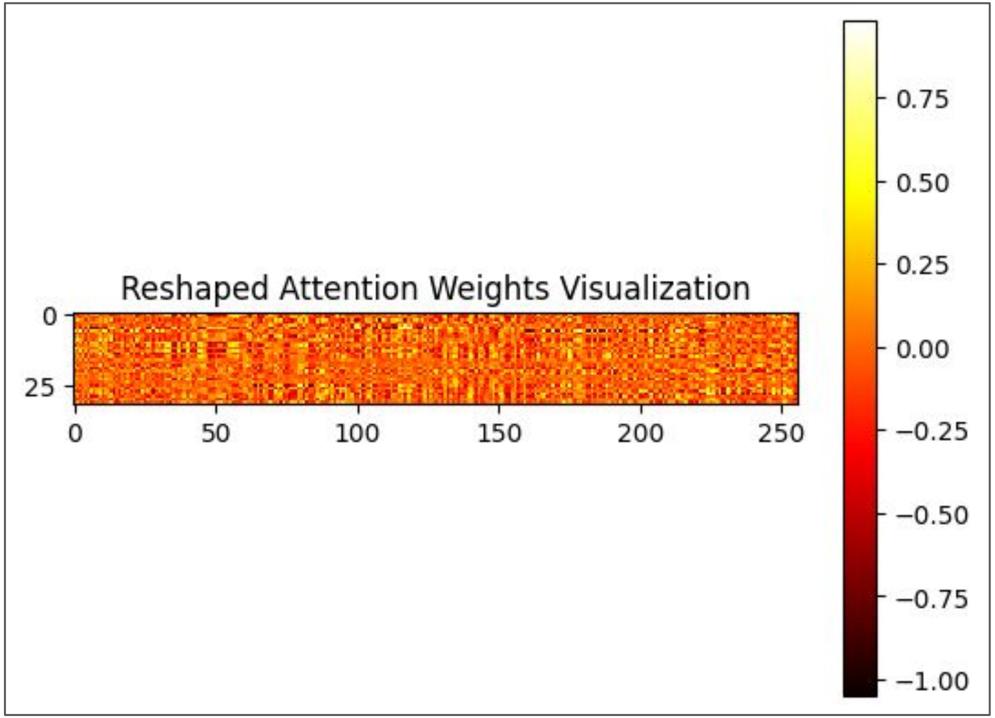


Figure 11: Reshaped Attention Weights Visualization

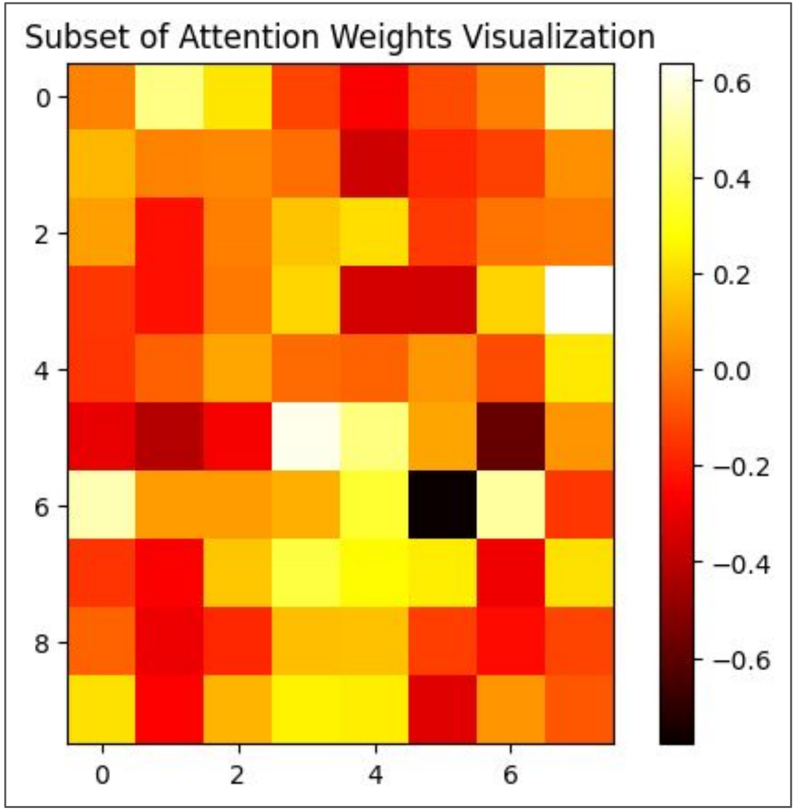


Figure 12: Subset Attention Weights Visualization

Results and Discussion

Class	Precision	Recall	f1-score	Support
0	0.97	0.98	0.97	20482
1	1.00	0.99	0.99	20665
2	0.99	1.00	0.99	20681
3	0.99	0.98	0.98	20768
Accuracy	0.99	0.98	0.99	82596
Macro avg	0.99	0.99	0.99	82596
Weighted avg	0.99	0.99	0.99	82596

Table: Statistical Report of Propose Model

Results and Discussion

Class	Precision	Recall	f1-score	Support
0	0.99	0.99	0.99	20482
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Table: Statistical Report of Classification

Results and Discussion

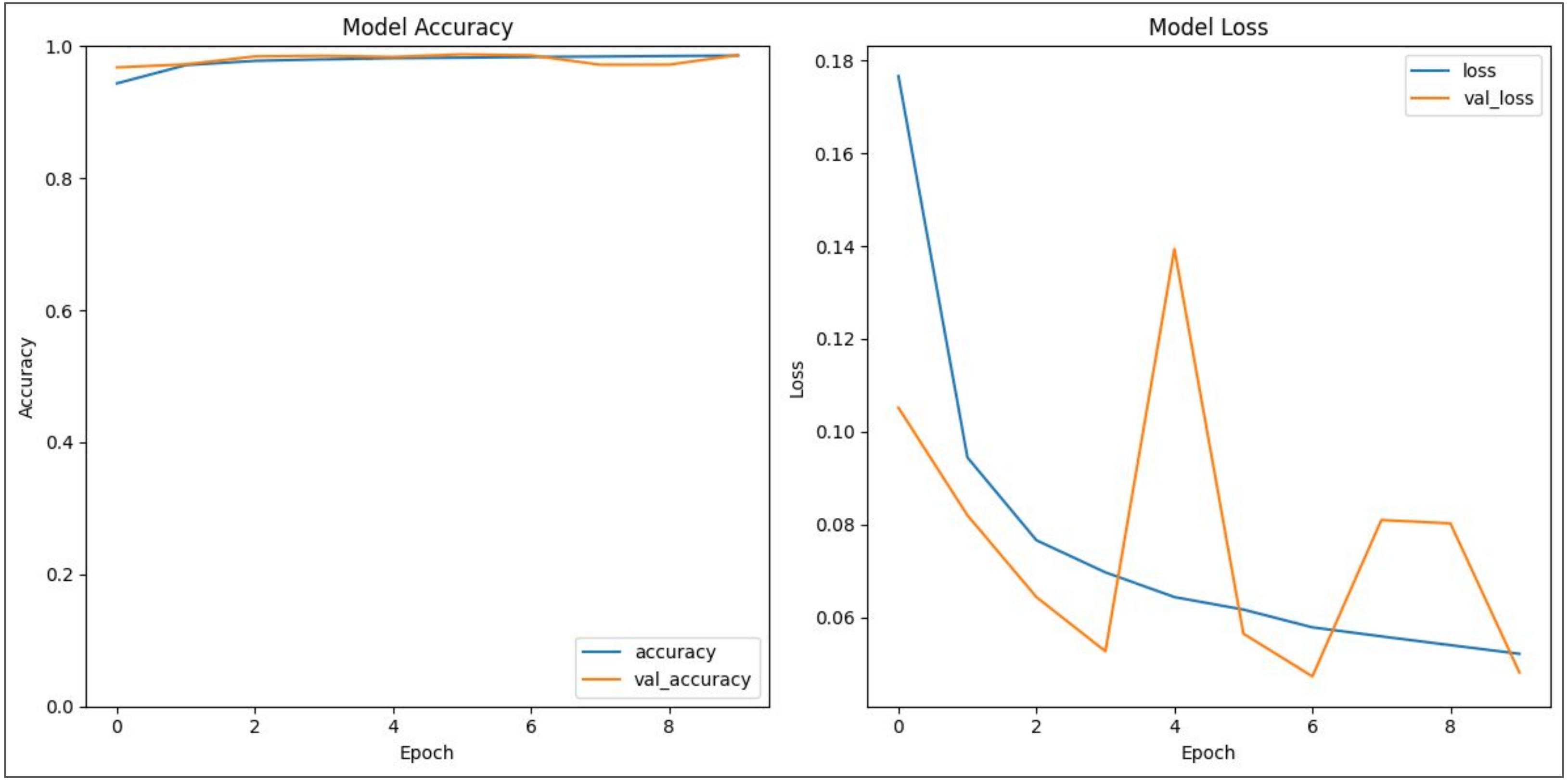

Figure 13: Model Accuracy and Loss Plotting

Competitive results

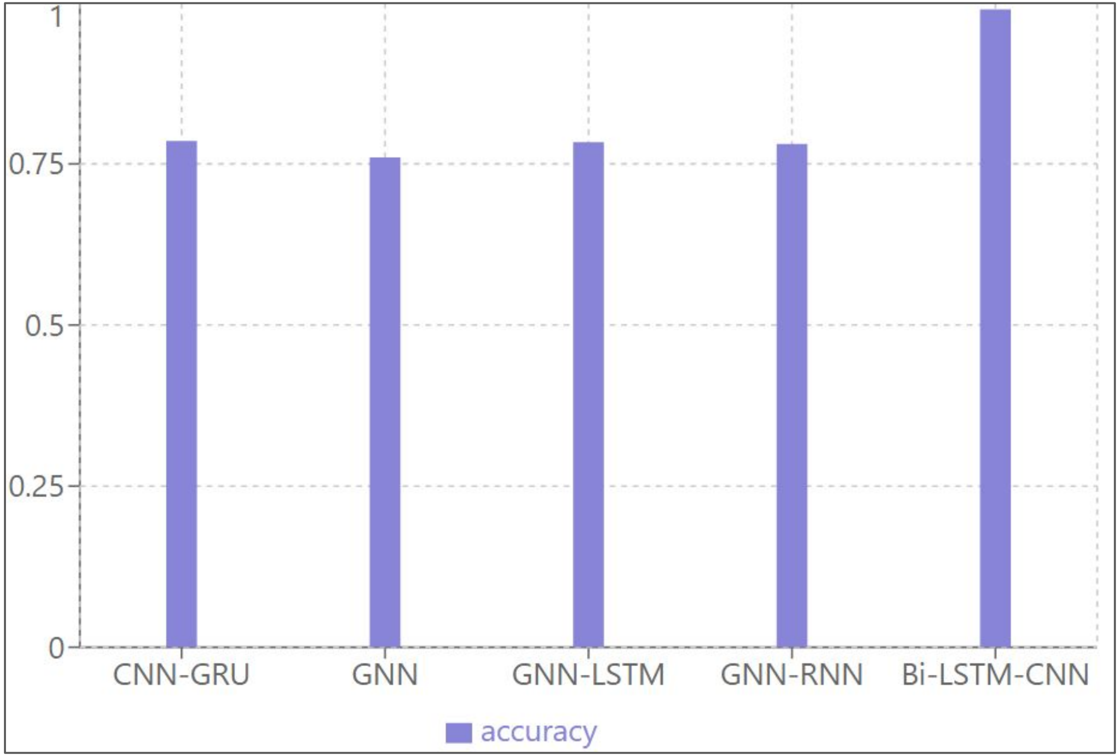

Figure 14: Performance Chart of Other Models Accuracy

Conclusion

The proposed **Bi-LSTM + CNN + Attention Layer hybrid model** effectively enhances intrusion detection by combining spatial feature extraction from CNN with temporal sequence analysis from Bi-LSTM. It addresses previous limitations by improving detection of complex attacks, reducing false positives, and offering greater adaptability to evolving threats. While highly effective, integrating an attention mechanism could further optimize focus on critical patterns for even better performance.

Future work

1. **Computational Complexity**: The hybrid model's combination of CNN and Bi-LSTM increases computational costs. Optimizing the model's architecture for **real-time intrusion detection** on resource-constrained systems would be beneficial.
2. **Limited Testing on Diverse Datasets**: The model has been tested on specific datasets. Expanding its evaluation to more diverse or real-world network datasets could ensure its **generalizability** across various network environments.(Web Application Specific Attacks e.g XSS, SQLIjection etc..)
3. **Scalability for Large Networks**: As networks grow in size and complexity, the model may face scalability challenges. Future improvements could focus on making the model more **scalable** and efficient for larger and more complex network infrastructures.

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Thank You